Dipok Deb

3017 White Ash Trail, Orlando, Florida, 32826
Mobile: +1(956)-240-0091
Email: dipok.deb@ucf.edu
LinkedIn: Dipok Deb

Professional Summary

Graduate student in Big Data Analytics with a strong foundation in data science, machine learning, deep learning, and statistical analysis. Experienced in formulating data-driven solutions, data exploration, and iterative model improvement. Skilled in Python, R, SAS, and ML/DL techniques applied to real-world problems like DDoS detection. Adept at collaborating with stakeholders to produce actionable insights and enhance customer experiences.

Research Interest

Machine Learning, Deep Learning, LLM, Cyber Security, Network Security, Quantum Cybersecurity, Quantum Computing, IoT Security, Data Science, Artificial Intelligence,

Technical Skills

- **Programming Languages:** Python, R, SAS, Fortran, MATLAB, C, Mathematica
- **Python Libraries:** PyTorch, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
- **Writing Tools:** MS Office, LaTeX
- **Operating Systems:** Ubuntu, Windows
- **General Skills:** Teamwork, Leadership Skills, Interpersonal Communication, Strong Mathematical Skills

Education

- **PhD in Big Data Analytics**
  University of Central Florida, Orlando, Florida, USA
  August 2024 - Present, CGPA: 3.83/4.0
- **Master of Science in Mathematics (Statistics)**
  University of Texas Rio Grande Valley, Edinburg, Texas, USA
  August 2022 – July 2024, CGPA: 3.84/4.0
- **Master of Science in Mathematics**
  Shahjalal University of Science & Technology, Bangladesh
  July 2015 - August 2017, CGPA: 3.82/4.0
- **Bachelor of Science in Mathematics**
  Shahjalal University of Science & Technology, Bangladesh
  January 2010 - June 2015, CGPA: 3.66/4.0

Employment History

- **Research Assistant, UCF**
  August 2026 - Present
- **Teaching Assistant, UCF**
  August 2024 - July 2026
  Taught Statistical Methods, Statistics Data Science AI-I, proctoring, grading, and maintaining office hours.
- **Research Assistant, UTRGV**
  June 2023 - July 2024
  Conducted research on DDoS attack detection using Machine Learning and Deep Learning tools.
- **Teaching Assistant, UTRGV**
  August 2022 - May 2023
  Taught Calculus II, Calculus III, Linear Algebra; prepared lectures, proctoring, grading.
- **Lecturer, SUST, Bangladesh**
  December 2018 - August 2022
  Taught various mathematics courses and conducted research in Applied Mathematics.

Relevant Courses at Graduate Level

Applied Time Series Analysis (STA 6857), Statistical Learning Theory (STA 7722), Current Topics in Big Data Analytics (STA 7935), Statistical Computing II (STA 6107), Bayesian Modelling Computation (STA 7348), Statistical Computing I (STA 6106), Parallel and Cloud Computation (COP 6526), Data Mining Methodology I (STA 5703), Data Science II (STA 6367), Data Science I (STA 6366), Theoretical Statistics I (STA 6326), Theoretical Statistics II (STA 6327), Linear Models (STA 6246), Regression Analysis (6236), Deep Learning and Neural Networks (CSCI 6379), Statistical Methods (MATH 6364), Algebra (MATH 6331), Biostatistics (STAT 6384), Stochastic Process (STAT 6379), Analysis (MATH 6352), Linear Algebra (MATH 6330), Statistical Learning (MATH 6333), Probability and Statistics (MATH 6365), Numerical Analysis (MATH 6375).

Research & Projects

- **DDoS Detection Using Machine Learning and Deep Learning** (Fall 2023 – Summer 2024)
  MSc thesis research focusing on enhancing DDoS detection using ML/DL techniques. MSc. Thesis.
- **Intrusion Detection System Comparison** (Spring 2024)
  Developed an IDS using FFNN, LSTM, and RandNN to assess intrusion detection in IoT networks. Poster.
- **Constraint Quantization for Probability Distribution** (Summer 2023)
  Research in quantization theory, obtaining optimal sets and errors for uniform distribution.

Publications

1. Beyond Random Splitting: Temporal Validation of Machine Learning for Chronic Kidney Disease Prediction, (Submitted, Under Review)
2. Topological Characterization of Global Physiological Biomarkers Dysregulation in Older Adults Among Type II Diabetic, (Submitted, Under Review)
3. Cross-Domain Evaluation of Machine Learning Models with Data Balancing Techniques for TCP SYN Flood DDoS Detection, (Submitted, Under Review)
4. Counting Household-Level Child Malnutrition Among Children Age Under Five Years in Bangladesh: A Spatial Cross-Sectional Study, (Submitted, Under Review)
5. Comparative Evaluation of Machine Learning Models with Different Data Balancing Techniques for DDoS Attack Detection, Preprint
6. Application and Analysis of Machine Learning and Deep Learning Algorithms in Detection of DDoS Cyberattacks, MSc. Thesis.
7. Constrained Quantization for a Uniform Distribution, ArXiv.
8. The Effects of Equivalence Ratio on Pressure Wave Development During Knocking Combustion, *Journal of Scientific Research*, 2018, Paper.

Conference Proceedings

1. Performance Analysis of ML Algorithms on Imbalanced DDoS Attack Dataset, 2024 IEEE World AI IoT Congress, Seattle, USA, Paper.
2. Constrained Quantization for a Uniform Distribution, AMS Fall Central Meeting, Omaha, Nebraska, 2023.

Grants & Awards

- VoloRidge Fellowship, School of Data, Mathematical, and Statistical Sciences (SDMSS), UCF, 2026
- Dean's Graduate Assistantship (DGA), UTRGV, 2022 – 2024
- Research Grants, SUST Research Centre, 2020 - 2022
- NST Fellowship, MS Thesis, 2016
- Academic Scholarship, SUST, 2011 - 2015

Professional Affiliations & Service

- Reviewer, Conference on Neural Information Processing Systems (NeurIPS), 2026.
- Member, IEEE Student, Member ID: 99400927, 2024 - Present.
- Member, American Mathematical Society (UTRGV Student Chapter), Member ID: DBDPXC, 2022-2024.
- Member, SPIE (UTRGV Student Chapter), 2022-2024.
- Executive Member, SUST Mathematics Alumni Association, 2016-Present.

Volunteer Activities

- Volunteer, National Undergraduate Mathematics Olympiad, Sylhet Region, 2012 - 2015.